

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Neurocore / GeoNavX (Location Intelligence & Robotics)

Challenge ID: PS-10

Title: 3D Autonomous Path Planning in GPS-Denied Environments

1. Executive Summary & Problem Overview

Autonomous robotic agents, unmanned aerial vehicles (UAVs / drones), and unmanned ground vehicles (UGVs) are increasingly deployed in high-risk operational environments where Global Positioning System (GPS) satellite signals are completely degraded, jammed, spoofed, or physically unreachable. These critical operating theaters include subterranean mining tunnels, collapsed disaster rubble, dense indoor warehouses, multi-level building interiors, underwater pipelines, and urban canyons.

In the absence of reliable external satellite positioning, traditional waypoint-based navigation algorithms fail catastrophically. Autonomous platforms must rely entirely on local onboard perception—such as monocular/stereo cameras, LiDAR, depth sensors, and Inertial Measurement Units (IMUs)—to perceive dynamic obstacles, build local spatial representations, and compute collision-free, kinematically feasible trajectories in real-time three-dimensional (3D) space.

The objective of **PS-10 (3D Autonomous Path Planning in GPS-Denied Environments)** is to design, develop, and benchmark a lightweight, computationally efficient, and intelligent 3D autonomous path planning and obstacle-avoidance algorithm. The system must fuse local positional odometry with environmental visual or sensor feeds to generate smooth, energy-efficient, and collision-free flight/traversal paths in complex, unknown GPS-denied environments within an interactive 3D simulation sandbox or software emulator.

2. Core Personas & Access Roles

1. **Autonomous Systems / Robotics Engineer:** Configures path-planning algorithms, tunes kinematic constraints (max velocity, acceleration, turning radius), and evaluates computational benchmark metrics (latency, path smoothness, memory footprint).
2. **Disaster Search & Rescue Coordinator:** Deploys virtual UAV agents into simulated collapsed structures, defining exploration goals, inspecting structural integrity, and locating trapped survivor coordinates.

3. **Subterranean / Industrial Inspector:** Manages autonomous drone inspections within GPS-deprived environments (e.g., HVAC shafts, utility tunnels, chemical silos, mining chambers).
4. **Mission Operations Specialist:** Monitors live 3D visual telemetry, vehicle state estimation, spatial point-cloud maps, and algorithm replanning triggers via a mission control console.

3. Key Functional Modules & Technical Deliverables

Module 1: 3D Environmental Representation & Voxel Mapping

- **Spatial Voxelization & Occupancy Grids:** Ingests simulated depth maps, point clouds, or LiDAR feeds to construct real-time 3D spatial representations (e.g., OctoMap, ESDF - Euclidean Signed Distance Fields, Voxel Grids).
- **Dynamic & Static Obstacle Differentiation:** Classifies immovable structural barriers (walls, pillars, ceilings) versus dynamic moving hazards (falling debris, moving machinery, other autonomous agents).
- **Lightweight Memory Management:** Employs spatial pruning and local rolling window maps to minimize memory footprint on resource-constrained embedded companion computers.

Module 2: State Estimation & Odometry Fusion (GPS-Denied)

- **Visual-Inertial Odometry (VIO) Simulation:** Estimates relative 6-DoF (Degrees of Freedom) vehicle pose (x, y, z, roll, pitch, yaw) by fusing IMU acceleration/gyroscope telemetry with optical flow or camera landmark tracking.
- **Drift Mitigation & Loop Closure:** Incorporates landmark graph optimization or keyframe matching to minimize cumulative dead-reckoning drift over prolonged traversals.
- **Fail-Safe Loss Detection:** Triggers emergency hover, station-keeping, or reverse-path backtracking when visual feature tracking degrades (e.g., low-light or featureless corridors).

Module 3: Intelligent 3D Path Planning & Optimization Engine

- **Global & Local Trajectory Planning:** Combines hierarchical path planners—such as 3D A*, Dijkstra, Jump Point Search, Rapidly-exploring Random Trees (RRT* / Informed RRT*), or Fast-Planner / Kinodynamic search—to find optimal collision-free paths.
- **Dynamic Real-Time Replanning:** Continually computes trajectory adjustments (<50ms latency) upon detecting sudden unmapped obstacles obstructing the flight corridor.
- **Trajectory Smoothing & Kinodynamic Feasibility:** Employs B-splines, minimum-snap optimization, or polynomial trajectory generation to ensure paths respect physical robotic limits (motor thrust, tilt angles, smooth cornering).

Module 4: 3D Mission Control Console & Benchmark Telemetry

- **Interactive 3D WebGL / Canvas Viewer:** Interactive 3D visualization dashboard (Three.js / Babylon.js / ROS Web / PyVista) rendering the environment, agent trajectory, sensor frustum, and planned path in real time.
- **Algorithmic Benchmark Analytics:** Computes comparative efficiency metrics: Total Path Length, Execution Runtime, Path Smoothness, Clearance Distance to Obstacles, and Computational CPU/RAM Overhead.
- **Scenario Customization Sandbox:** Allows users to spawn various complex GPS-denied environments (e.g., subterranean labyrinth, warehouse racks, indoor office, forest canopy) with configurable start/goal coordinates.

4. Recommended Technology Stack

Layer, Suggested Technologies

3D Visualization & Frontend, React.js (Vite) / Next.js / Tailwind CSS / Three.js / React Three Fiber / Babylon.js

Backend & Algorithm Engine, Python (FastAPI / WebSockets) or C++ (wrapped via pybind11 / WebAssembly)

Robotics & Simulation Tools, "ROS / ROS2 (Robot Operating System), Gazebo, Webots, AirSim, PyBullet, or Custom Python 3D Simulator"

Path Planning & Math, "Python (NumPy, SciPy, Open3D, Shapely, NetworkX), OMPL (Open Motion Planning Library)"

State Estimation & Vision, "OpenCV (Visual Odometry simulation), PyEKF (Extended Kalman Filtering)"

Deployment & Container, Dockerized environment or WebAssembly (Wasm) interactive web deployment

5. Evaluation & Judging Criteria (100 Points Total)

1. **Algorithmic Rigor & Path Optimality (25 Points):** Correctness, computational efficiency, and mathematical soundness of the 3D path planning and trajectory optimization algorithms.
2. **Obstacle Avoidance & Dynamic Replanning (25 Points):** Robust real-time avoidance of static and dynamic 3D obstacles with low-latency replanning loops.
3. **GPS-Denied Odometry & State Simulation (20 Points):** Realistic handling of positional estimation using camera, IMU, or sensor fusion without relying on global GPS coordinates.
4. **3D Visualization & Mission Console UX (15 Points):** High-performance interactive 3D rendering, clear path tracing, responsive camera controls, and intuitive scenario generation.
5. **Software Prototype, Code Standards & Benchmarking (15 Points):** Clean GitHub architecture, reproducible simulation runner, .env.example, documented algorithmic metrics, and a working live demo.

6. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup guide, simulator launch scripts, and .env.example.
2. **Live Deployment / Demonstration:** Interactive hosted web simulator (Vercel / Render / Hugging Face Spaces) or a reproducible Docker/screen demonstration.
3. **Demo Video / Deck:** A crisp 3-minute video showing:
 4. Initialization of a complex 3D GPS-denied test environment (e.g., tunnel or multi-level building).
 5. Setting start and goal waypoints without GPS data.
 6. Real-time 3D voxel mapping and collision-free path planning computation.
 7. Spawning a sudden unexpected obstacle and observing autonomous dynamic trajectory replanning.
 8. Performance telemetry dashboard displaying path metrics and compute latency.